

IMPERIAL



Recent changes in the Firedrake software ecosystem

Connor Ward
16/09/2025

- Firedrake '22** pyop3: A new domain-specific language for automating high-performance mesh-based simulation codes
- Firedrake '23** Latest developments in pyop3
- Firedrake '24** pyop3 is coming
- Firedrake USA '25** pyop3: a better code generator for mesh-based computations



Submit thesis/Begin postdoc

10/24



New Mac runners

11/24



Absorb PyOP2 and TSFC in the Firedrake repository

12/24



New Linux runners

02/25

2 of:

- 384GB RAM
- NVIDIA A100 GPU
- 4 runners



pip install firedrake!

02/25



How did we get here?

- Firedrake needs a particular environment of compiled dependencies
- Most users are unfamiliar with compiling complex software
- HPC makes this even harder
- Hence `firedrake-install`



How did we get here?

- Firedrake needs a particular environment of compiled dependencies
- Most users are unfamiliar with compiling complex software
- HPC makes this even harder
- Hence `firedrake-install`

But.



How did we get here?

- Firedrake needs a particular environment of compiled dependencies
- Most users are unfamiliar with compiling complex software
- HPC makes this even harder
- Hence `firedrake-install`

But.

- `firedrake-install` was not a robust solution
- Install failures were extremely hard to debug



What happened?

- `pip install libsupermesh`
- `pip install firedrake`
- 3 stage install with `firedrake-configure` (intentionally stupid)
- OpenMPI instead of MPICH (`firedrake-run-split-tests`)
- `firedrake-check`
- `--netgen` etc into `pyproject.toml`
- New Docker images



Success stories

- Much easier HPC install
- Other packaging systems (PETSc, NixOS)



Smoke tests

02/25

- Gusto
- Thetis
- G-ADOPT
- Spyro



Smoke tests

02/25

- Gusto
- Thetis
- G-ADOPT
- Spyro

It really works!



How to add smoke tests for your library

```
- name: Run Gusto smoke tests
  # Only test Gusto in real mode
  if: |
    inputs.run_tests &&
    (success() || steps.install.conclusion == 'success') &&
    matrix.arch == 'default'
  run: |
    . venv/bin/activate
    git clone --depth 1 https://github.com/firedrakeproject/gusto.git gusto-repo
    pip install --verbose ./gusto-repo
    python -m pytest -n 8 --verbose \
      gusto-repo/integration-tests/balance/test_saturated_balance.py \
      gusto-repo/integration-tests/equations/test_thermal_sw.py \
      gusto-repo/integration-tests/transport/test_embedded_dg_advection.py
    timeout-minutes: 10
```



02/25



mpi-pytest

```
def test_comm_world_size_equals_two():  
    assert COMM_WORLD.size == 2  
  
def test_comm_world_size_equals_three():  
    assert COMM_WORLD.size == 3
```

\$ pytest test_comms.py *# won't work!*

\$ mpiexec -n 2 pytest test_comms.py *# won't work!*



mpi-pytest

```
@pytest.mark.parallel(2) $ pytest test_comms.py # works, calls MPI under the hood
def test_comm_world_size_equals_two():
    assert COMM_WORLD.size == 2 $ mpiexec -n 2 pytest test_comms.py -m parallel[2] # works
```

```
@pytest.mark.parallel(3)
def test_comm_world_size_equals_three():
    assert COMM_WORLD.size == 3
```



mpi-pytest

```
@pytest.mark.parallel(2) $ pytest test_comms.py # works, calls MPI under the hood
def test_comm_world_size_equals_two():
    assert COMM_WORLD.size == 2 $ mpiexec -n 2 pytest test_comms.py -m parallel[2] # works
```

```
@pytest.mark.parallel(3)
def test_comm_world_size_equals_three():
    assert COMM_WORLD.size == 3
```

`pip install mpi-pytest`



Stop using our own fork of PETSc

03/25



First Firedrake release!

04/25



Why it happened

- To provide more stability for downstream packages



How it happened

1. Coordinated the release of UFL, pyadjoint, loopy
2. Made the first releases of FIAT and libsupermesh
3. 2025.4.0 released on PyPI



How it happened

1. Coordinated the release of UFL, pyadjoint, loopy
2. Made the first releases of FIAT and libsupermesh
3. 2025.4.0 released on PyPI
4. It immediately broke
5. 2025.4.0.post0 released



Understanding Firedrake's release strategy



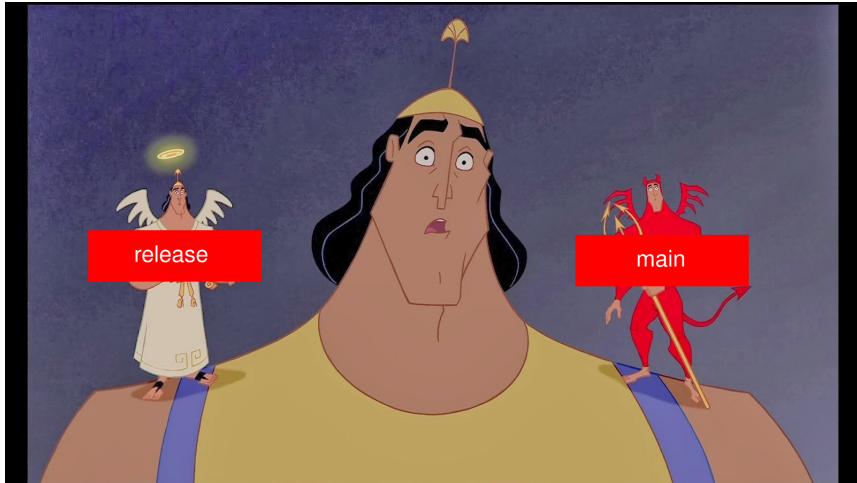
Understanding Firedrake's release strategy



Understanding Firedrake's release strategy



Understanding Firedrake's release strategy



Understanding Firedrake's release strategy

2025.4.0 release

main



Understanding Firedrake's release strategy

	docs	breaking change 1
	bugfix 1	new feature 1
2025.4.0	release	main

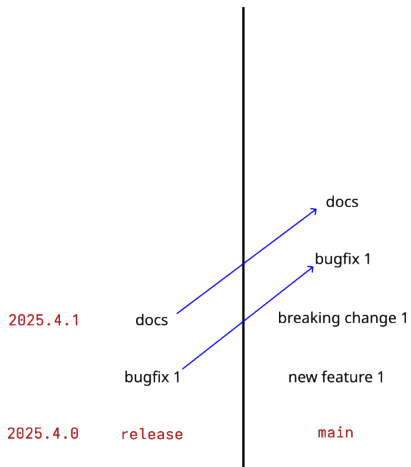


Understanding Firedrake's release strategy

2025.4.1	docs	breaking change 1
	bugfix 1	new feature 1
2025.4.0	release	main



Understanding Firedrake's release strategy



Understanding Firedrake's release strategy

2025.4.1

docs

bugfix 1

2025.4.0

release

docs

bugfix 1

breaking change 1

new feature 1

main



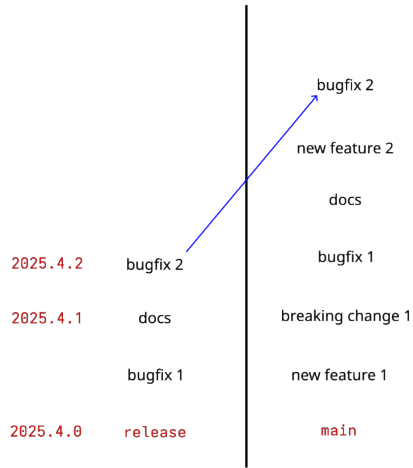
Understanding Firedrake's release strategy



Understanding Firedrake's release strategy



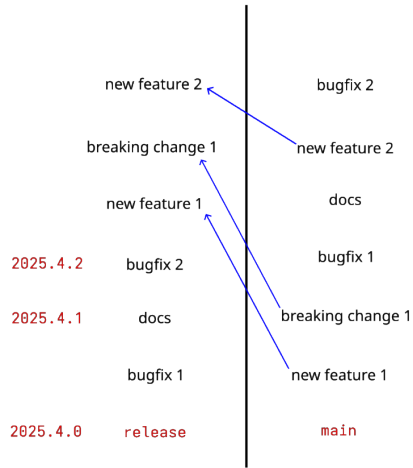
Understanding Firedrake's release strategy



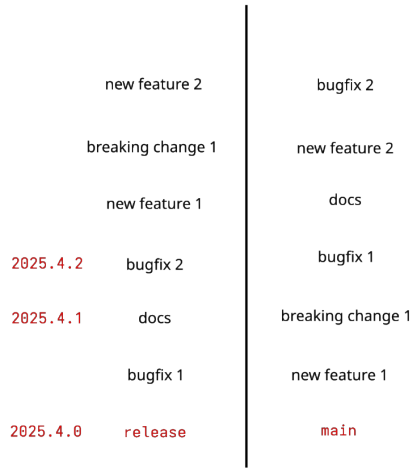
Understanding Firedrake's release strategy



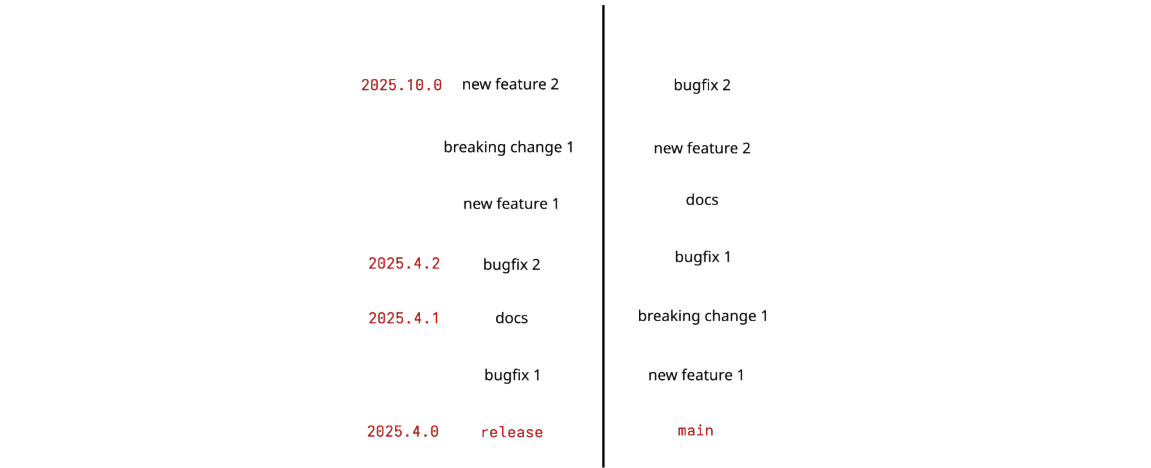
Understanding Firedrake's release strategy



Understanding Firedrake's release strategy



Understanding Firedrake's release strategy



Supporting main and/or release



Supporting main and/or release

Strategy 1 Just pick one.



Supporting main and/or release

Strategy 1 Just pick one.

Strategy 2 Maintain your own release and main branches.



Supporting main and/or release

Strategy 1 Just pick one.

Strategy 2 Maintain your own release and main branches.

Strategy 3 Check the version.

```
if importlib.metadata.version("firedrake") > "2025.4":  
    # new API  
else:  
    # old API
```



Supporting main and/or release

Strategy 1 Just pick one.

Strategy 2 Maintain your own release and main branches.

Strategy 3 Check the version.

```
if importlib.metadata.version("firedrake") > "2025.4":  
    # new API  
else:  
    # old API
```

We have Docker images for release (firedrake-vanilla-default:dev-release) and main (firedrake-vanilla-default:dev-main) that you can use for your CI.



Use more system packages when building PETSc

04/25

From:

Depending on your platform, PETSc may take an hour or more to build!

To:

- | | | | |
|---|---|---------------------------------|--------|
| > | ✓ | Install system dependencies (2) | 37s |
| > | ✓ | Install PETSc | 2m 21s |
| > | ✓ | Install Firedrake | 9m 0s |



07/25



Motivation

The screenshot shows a GitHub pull request interface. At the top, the repository is 'dolphin-adjoint / pyadjoint'. The pull request title is 'Use petsctools #214'. It shows that 'jrmaddison' wants to merge 1 commit into 'dolphin-adjoint:master' from 'jrmaddison:jrmaddison/petsctools'. The pull request has 19 issues, 9 pull requests, and 1 action. The conversation section shows a comment from 'jrmaddison' stating 'No description provided.' with 1 reaction. The reviewers section shows 'connorjward' as the reviewer. The code diff shows a modified version of the 'flatten_parameters' function from 'firedrake/petsc.py'.

Use petsctools #214

[Open](#) jrmaddison wants to merge 1 commit into `dolphin-adjoint:master` from `jrmaddison:jrmaddison/petsctools`

Conversation 1 Commits 1 Checks 0 Files changed 1 +7 -200

jrmaddison commented yesterday

Member

No description provided.

1

```
130 - # Modified version of flatten_parameters function from firedrake/petsc.py,
131 - # Firedrake master branch 57e21cc8ebdb044c1d8423b48f3dbf70975d5548, first
132 - # added 2024-08-08
133 - def flatten_parameters(parameters, sep="_"):
134 -     """Flatten a nested parameters dict, joining keys with sep.
```



Features

- Interacting with PETSc options: `OptionsManager` (think `solver_parameters`)
- Passing arbitrary data through PETSc to Python preconditioners: 'app context'
- Handling citations
- Help initialising PETSc
- Reading PETSc configuration information



Features

- Interacting with PETSc options: `OptionsManager` (think `solver_parameters`)
- Passing arbitrary data through PETSc to Python preconditioners: 'app context'
- Handling citations
- Help initialising PETSc
- Reading PETSc configuration information

`pip install petsctools`

<https://firedrakeproject.org/petsctools>



Final remarks

Now

- A **lot** has changed in Firedrake in the past year
- 2025.10.0 is coming soon
- Feedback welcome on what the community wants next (`pip install petsc?`, `conda?`, `spack?`)
- I am very happy to help people who want to try any of this out for their own project

